Krishiv Singhal

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Education

Carnegie Mellon University *Aug 2024–Dec 2027*
Major in Mathematics and Logic and Computation *Pittsburgh, PA*

- Notable coursework: Advanced NLP (PhD), Machine Learning (PhD), Cooperative AI (Graduate), Modal Logic (Graduate), Formal Logic, Probability Theory, Linear Algebra, Abstract Algebra, Real and Complex Analysis. GPA: 3.7.
- Studied abroad at the University of Sydney; completed advanced coursework in maths on a full merit scholarship; awarded the Dalyell Scholarship and Chancellor's Award. GPA: Distinction.

United World College, Singapore *Aug 2017–May 2024*
International Baccalaureate *Singapore*

- Perfect IB Score: 45/45 (top 0.5% worldwide); ranked first in class for thesis on the Curry–Howard correspondence and intuitionistic logic.

Industry Experience

Quantitative Developer Intern *Jun–Aug 2026*
Stealth Systematic Macro Fund; reference available on request *Hong Kong*

- Built a Bayesian regime estimator for global equity indices and adopted it into the fund's research stack. Monthly ranking strategies run on it lifted Sharpe by 0.05–0.15 versus the cross-section, cutting participation by up to 75%.
- Engineered co-activity and regime-flow analytics tracking probability mass across market states, surfacing churn conditions.

Selected Projects

PlainSight — Australian Legal AI and Contract Retrieval *Python, LangChain, RAG*

- Built an end-to-end retrieval-augmented generation prototype for Australian contract and case-law analysis: ingested and chunked legal corpora, embedded documents into a custom vector store, and grounded LLM responses in retrieved authorities.
- Prototyped clause-level analysis for commercial agreements and a hybrid LLM plus structured-rules architecture for jurisdiction-specific risk detection, explainable compliance checks, and suggested revisions.

Llama-Style Language Model *PyTorch*

- Implemented a 25M-parameter Llama-style transformer from scratch, including BPE tokenization, rotary embeddings, grouped-query attention, RMSNorm, SwiGLU, causal masking, and AdamW; trained on a custom 212M-token literary and philosophical corpus, with controlled upsampling experiments across selected authors.

Bitewise Labs — Bites Suite *Flutter, Swift*

- Founded an iOS studio and shipped three cultural-education apps (Literature, Art, and Architecture Bites): about 1,800 downloads and consistent revenue, across 2,323 quotations, 985 artworks, and 641 buildings. Built a short-form video pipeline.

Differentiable Kripke Semantics for Modal Model Search *PyTorch, NumPy*

- Implemented differentiable fuzzy Kripke semantics and formulated modal satisfiability as gradient-based minimax optimization over accessibility relations and adversarial valuations; extending the approach to modal constraints for RL policies.

Research Experience

AI Evaluation Research Group *Sep 2026–Present*
Carnegie Mellon University

- Selected to Prof Sarah Cen's research group on prediction and recalibration of model performance under covariate shift, including synthetic distribution-shift experiments and unlabeled recalibration methods.

Probabilistic Modal Logic and Formal Verification *Mar 2026–Present*
Research, Carnegie Mellon University

- Developing graded probabilistic semantics for propositional dynamic logic, deriving quantitative interpretations of classical modal axioms over stochastic transition systems and exploring applications to formal safety specifications for AI agents. Research under Prof. Adam Bjorndahl.

Awards

Chancellor's Scholarship, the highest undergraduate award at the University of Sydney · **UWC Global Academy,** fully funded selective residential program at the University of Chicago · **Canadian Senior Maths Contest,** International Honour Roll, global top 1% · **SASMO Gold,** Global top 200. · **AMC10,** AIME Qualifier, global top 2.5%. · **AIME:** Scored 8/15.

Skills and Interests

Languages: Python, Swift, SQL, LaTeX · **AI / ML:** PyTorch, NumPy, pandas, LangChain, RAG, LLM APIs
Also: Flutter, Lean 4, Isabelle.

Interests: Music production (composed, produced, and released 5 albums with 40,000 streams on major streaming platforms; remix favourite songs on YouTube); golf; poker.